MTH 204: Lecture 21

Application to Nonhomogeneous Linear ODEs:

Let us consider $y'' + ay' + by = r(t)$

where a and b are constants

Taking Laplace Transform we get,

$$s^2 Y(s) - sy(0) - y'(0) + a\{sY(s) - y(0)\} + bY(s) = R(s)$$

$$\Rightarrow \quad (s^2 + as + b)Y(s) = (s+a)y(0) + y'(0) + R(s)$$

$$\Rightarrow \quad Y(s) = \frac{(s+a)y(0) + y'(0) + R(s)}{(s^2 + as + b)}$$

$$\Rightarrow \quad Y(s) = \frac{(s+a)y(0) + y'(0)}{(s^2 + as + b)} + \frac{R(s)}{(s^2 + as + b)}$$

Now if $\quad Q(s) = \dfrac{1}{(s^2 + as + b)}$

then $\quad Y(s) = \{(s+a)y(0) + y'(0)\}Q(s) + R(s)\,Q(s)$

If $y(0) = 0$, $y'(0) = 0$ (Response of system at rest)

then $\quad Y(s) = R(s)\,Q(s)$

Hence $\quad y(t) = \displaystyle\int_0^t r(\tau)\,q(t-\tau)\,d\tau$

i.e. $\boxed{y(t) = r(t) * q(t)}$

Response of a Damped Vibrating System to a Square Wave:

$$y'' + 5y' + 6y = r(t) \quad \text{where} \quad r(t) = \begin{cases} 1 & \text{if } 0<t<a \\ 0 & \text{if } t>a \end{cases} \quad \text{and} \quad \begin{array}{l} y(0)=0 \\ y'(0)=0 \end{array}$$

After taking Laplace Transform the equation becomes $Y(s)(s^2+5s+6) = R(s)$

$$\Rightarrow Y(s) = \frac{1}{(s^2+5s+6)} R(s) = Q(s)R(s)$$

where $Q(s) = \dfrac{1}{s^2+5s+6} = \dfrac{1}{s+2} - \dfrac{1}{s+3}$

Then $q(t) = e^{-2t} - e^{-3t}$

So, $\quad y(t) = r(t) * q(t) = r(t) * \left(e^{-2t} - e^{-3t}\right)$

If we take the indefinite integral of

$$\int q(t-\tau) \times 1 \, d\tau = \int \left[e^{-2(t-\tau)} - e^{-3(t-\tau)} \, d\tau\right]$$

$$= \frac{1}{2} e^{-2(t-\tau)} - \frac{1}{3} e^{-3(t-\tau)}$$

Now <u>if $t<a$</u>, $\quad r(t) * q(t) = \int_0^t \left[e^{-2(t-\tau)} - e^{-3(t-\tau)}\right] d\tau$

$$= \left[\frac{1}{2} e^{-2(t-\tau)} - \frac{1}{3} e^{-3(t-\tau)}\right]_0^t$$

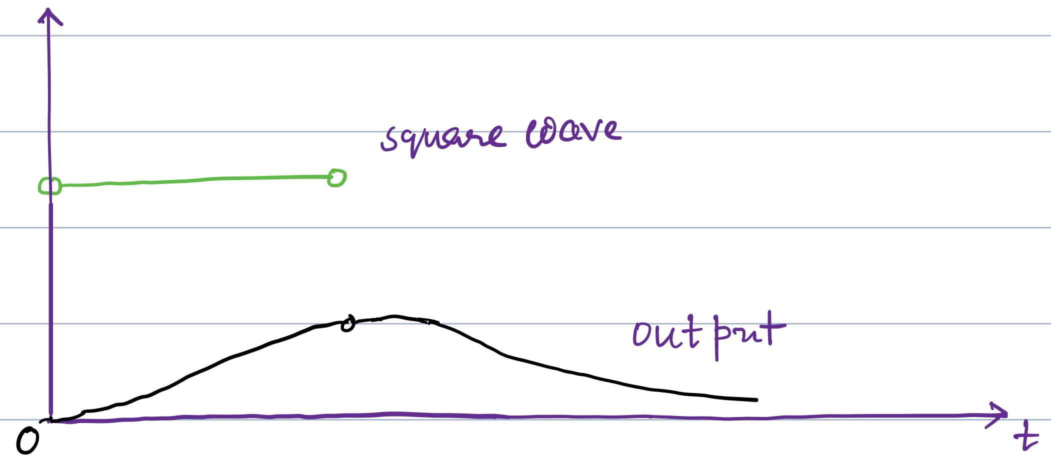
$$= \frac{1}{2} - \frac{1}{3} - \frac{1}{2} e^{-2t} + \frac{1}{3} e^{-3t}$$

$$= \boxed{\frac{1}{6} - \frac{1}{2} e^{-2t} + \frac{1}{3} e^{-3t}}$$

If $t > a$, $r(t) * q(t) = \int_0^a \left[e^{-2(t-\tau)} - e^{-3(t-\tau)} \right] d\tau$

$$= \left[\frac{1}{2} e^{-2(t-\tau)} - \frac{1}{3} e^{-3(t-\tau)} \right]_0^a$$

$$= \boxed{\frac{1}{2} e^{-2(t-a)} - \frac{1}{3} e^{-3(t-a)} - \frac{1}{2} e^{-2t} + \frac{1}{3} e^{-3t}}$$



A Voterra Integral Equation of the second kind :

$$y(t) + \int_0^t y(\tau) \cosh(t-\tau) d\tau = t + e^t$$

It can be written as

$$y(t) + y * \cosh(t) = t + e^t$$

Taking Laplace Transform we get

$$Y(s) + Y(s)\frac{s}{s^2-1} = \frac{1}{s^2} + \frac{1}{s-1}$$

$$\Rightarrow Y(s)\frac{(s^2-1+s)}{s^2-1} = \frac{1}{s^2} + \frac{1}{s-1}$$

$$\Rightarrow Y(s) = \frac{(s^2-1)}{\cancel{(s^2+s-1)}} \frac{\cancel{(s-1+s^2)}}{s^2(s-1)}$$

$$\Rightarrow Y(s) = \frac{(s+1)}{s^2} = \frac{1}{s} + \frac{1}{s^2}$$

Taking Inverse Laplace Transform, $\boxed{y(t) = 1+t}$

Ex: $$y(t) + \int_0^t e^{2(t-\tau)} y(\tau)\,d\tau = t^2 - t - \frac{1}{2} + \frac{1}{2}e^{2t}$$

We can write it as

$$y(t) + y(t) * e^{2t} = t^2 - t - \frac{1}{2} + \frac{1}{2}e^{2t}$$

Taking Laplace Transform:

$$Y(s) + Y(s)\frac{1}{s-2} = \frac{2}{s^3} - \frac{1}{s^2} - \frac{1}{2s} + \frac{1}{2(s-2)}$$

$$\Rightarrow Y(s)\left(\frac{s-2+1}{s-2}\right) = \frac{4(s-2) - 2s(s-2) - s^2(s-2) + s^3}{2s^3(s-2)}$$

$$\Rightarrow Y(s)\frac{(s-1)}{(s-2)} = \frac{4s - 8 - \cancel{2s^2} + 4s - \cancel{s^3} + \cancel{2s^2} + \cancel{s^3}}{2s^3(s-2)}$$

$$\Rightarrow \quad Y(s) = \frac{\cancel{(s-2)}}{\cancel{(s-1)}} \frac{8\cancel{(s-1)}}{2s^3\cancel{(s-2)}} = \frac{4}{s^3}$$

Taking Inverse Laplace Transform : $\boxed{y(t) = 2t^2}$

Differentiation of Transforms :

If f satisfies the assumptions of the existence theorem:

then $\mathcal{L}\{t f(t)\} = -F'(s)$ where $F(s) = \mathcal{L}\{f(t)\}$

Hence $\mathcal{L}^{-1}\{F'(s)\} = -t f(t)$

Proof:

$$F(s) = \mathcal{L}(f(t)) = \int_0^\infty e^{-st} f(t)\,dt$$

$$F'(s) = \frac{d\left\{\int_0^\infty e^{-st} f(t)\,dt\right\}}{ds}$$

$$= \int_0^\infty \frac{d}{ds}\left\{e^{-ts}\right\} f(t)\,dt$$

$$= \int_0^\infty -t\,e^{-st} f(t)\,dt$$

$$= \int_0^\infty e^{-st}\left(-t f(t)\right) dt$$

$$= \mathcal{L}\{-t f(t)\}$$

So, $\mathcal{L}\{t f(t)\} = -F'(s)$

Hence $\mathcal{L}^{-1}(F'(s)) = -t f(t)$

Ex: $\mathcal{L}\{\sin(\omega t)\} = \dfrac{\omega}{s^2+\omega^2} = F(s)$

$$\mathcal{L}\{-t\sin(\omega t)\} = F'(s) = \frac{-2\omega s}{(s^2+\omega^2)^2}$$

$$\mathcal{L}\{-t\sin(\omega t)\} = -\frac{2\omega s}{(s^2+\omega^2)^2}$$

$$\mathcal{L}\{t\sin(\omega t)\} = \frac{2\omega s}{(s^2+\omega^2)^2}$$

$$\mathcal{L}^{-1}\left\{\frac{2\omega s}{(s^2+\omega^2)^2}\right\} = t\sin(\omega t)$$

Integration of Transforms:

If f satisfies the assumptions of the existence theorem and the limit of $\frac{f(t)}{t}$ exists when t approaches 0 from the right then $\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(x)\,dx$

Hence, $$\mathcal{L}^{-1}\left\{\int_s^\infty F(x)\,dx\right\} = \frac{f(t)}{t}$$

We know $$F(x) = \int_0^\infty e^{-xt} f(t)\,dt$$

$$\int_s^\infty F(x)\,dx = \int_s^\infty \left\{\int_0^\infty e^{-xt} f(t)\,dt\right\} dx$$

$$= \int_0^\infty \left\{\int_s^\infty e^{-xt} f(t)\,dx\right\} dt$$

$$= \int_0^\infty f(t) \left\{\int_s^\infty e^{-xt}\,dx\right\} dt$$

$$= \int_0^\infty f(t) \left[\frac{e^{-xt}}{-t}\right]_s^\infty dt$$

$$= \int_0^\infty f(t) \frac{e^{-st}}{t}\,dt = \int_0^\infty e^{-st} \frac{f(t)}{t}\,dt$$

$$= \mathcal{L}\left\{\frac{f(t)}{t}\right\}$$

$$\mathcal{L}^{-1}\left\{ \int_s^\infty F(x)\,dx \right\} = \frac{f(t)}{t}$$

<u>Ex:</u> Calculate $\mathcal{L}^{-1}\left\{ \log \frac{(s^2+w^2)}{s^2} \right\}$

Let us define $G = F' = \frac{d\left\{ \log\left(\frac{s^2+\omega^2}{s^2}\right) \right\}}{ds}$

$$= \frac{2s}{s^2+\omega^2} - \frac{2s}{s^2}$$

$$= \frac{2s}{s^2+\omega^2} - \frac{2}{s}$$

Then $g = \mathcal{L}^{-1}\left\{ \frac{2s}{s^2+\omega^2} - \frac{2}{s} \right\}$

$$= 2\cos(\omega t) - 2 = 2\left[\cos(\omega t) - 1\right].$$

$$\mathcal{L}^{-1}(F) = \mathcal{L}^{-1}\left\{ -\int_s^\infty G(s)\,ds \right\}$$

$$= -\frac{g(t)}{t} = \boxed{\frac{-2\left(\cos(\omega t) - 1\right)}{t}}$$

ODEs with Variable Coefficients

Let $\mathcal{L}(y) = Y$

$$\mathcal{L}\{y'\} = sY - y(0)$$

$$\mathcal{L}\{y''\} = s^2Y - sy(0) - y'(0)$$

$$\mathcal{L}\{t y'\} = -\frac{d\{sY - y(0)\}}{ds}$$

$$= -[Y + sY'] = -Y - sY'$$

$$\mathcal{L}\{ty''\} = -\frac{d\{s^2Y - sy(0) - y'(0)\}}{ds}$$

$$= -[2sY + s^2Y' - y(0)]$$

$$= -2sY - s^2Y' + y(0)$$

Ex: Laguerre's Equation: (Lah-GAIR) → (Pronounciation)

$$ty'' + (1-t)y' + ny = 0$$

Taking Laplace Transform,

$$-2sY - s^2Y' + y(0) + sY - Y(0) + Y + sY' + nY = 0$$

$$\Rightarrow (s-s^2)Y' + (n+1-s)Y = 0$$

$$\Rightarrow Y' = -\frac{(n+1-s)}{(s-s^2)}Y$$

$$\Rightarrow \frac{Y'}{Y} = \frac{(n+1-s)}{s(s-1)}$$

$$\Rightarrow \frac{dY}{Y} = \left(\frac{n}{s-1} - \frac{n+1}{s}\right)ds$$

$$\Rightarrow \log Y = n\log(s-1) - (n+1)\log(s)$$

$$\Rightarrow \log Y = \log\left\{\frac{(s-1)^n}{s^{n+1}}\right\}$$

$$\Rightarrow \boxed{Y = \frac{(s-1)^n}{s^{n+1}}}$$

Note that, $\mathcal{L}\{t^n e^{-t}\} = \dfrac{n!}{(s+1)^{n+1}}$ $\left(\text{since } \mathcal{L}(t^n) = \dfrac{n!}{s^{n+1}}\right)$

$$\Rightarrow \mathcal{L}\left\{\frac{d^n(t^n e^{-t})}{dt^n}\right\} = \frac{n!\, s^n}{(s+1)^n} \quad \left(\begin{array}{l}\text{derivatives upto order}\\ (n-1) \text{ are zero at } 0\end{array}\right)$$

$$\Rightarrow \mathcal{L}^{-1}\left\{\frac{s^n}{(s+1)^{n+1}}\right\} = \frac{1}{n!}\frac{d^n(t^n e^{-t})}{dt^n} \quad \text{.....(1)}$$

Let $L_0 = 1$, $\quad L_n(t) = \frac{e^t}{n!}\frac{d^n(t^n e^{-t})}{dt^n}$ for $n = 1, 2, \ldots$

These are polynomials because exponential terms cancel if we perform the indicated differentiations. They are called Lagurre Polynomials.

From ① $\mathcal{L}\left\{\frac{e^t}{n!}\frac{d^n(t^n e^{-t})}{dt^n}\right\} = \frac{(s-1)^n}{s^{n+1}}$

$$\Rightarrow \mathcal{L}^{-1}\left\{\frac{(s-1)^n}{s^{n+1}}\right\} = \frac{e^t}{n!}\frac{d^n(t^n e^{-t})}{dt^n} \Rightarrow \boxed{\mathcal{L}^{-1}(Y) = L_n(t)} \text{ i.e. } \boxed{y = L_n(t)}$$

Note: $L_n(t) = \frac{e^t}{n!}\frac{d^n(t^n e^{-t})}{dt^n} = \frac{1}{n!}(D-1)^n(t^n)$

Systems of ODEs:

$$\left.\begin{aligned} y_1' &= a_{11}y_1 + a_{12}y_2 + g_1(t) \\ y_2' &= a_{21}y_1 + a_{22}y_2 + g_2(t) \end{aligned}\right\}$$

Taking the Laplace Transform of both sides,

$$\left.\begin{aligned} sY_1 - y_1(0) &= a_{11}Y_1 + a_{12}Y_2 + G_1(s) \\ sY_2 - y_2(0) &= a_{21}Y_1 + a_{22}Y_2 + G_2(s) \end{aligned}\right\}$$

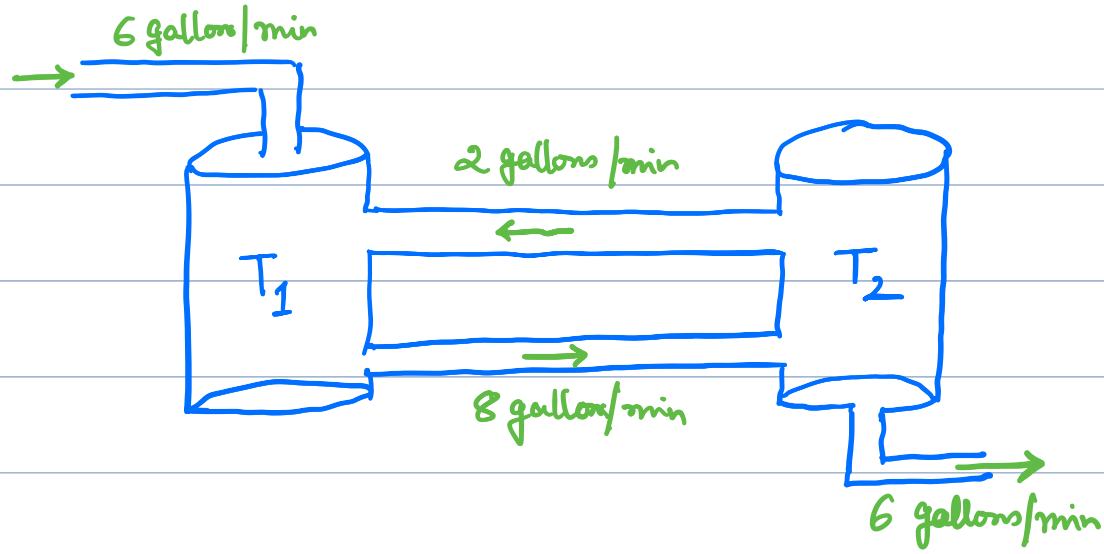
where $Y_i = \mathcal{L}(y_i)$, $G_i = \mathcal{L}(g_i)$ for $i = 1, 2$

Therefore
$$\left.\begin{aligned} (a_{11} - s)Y_1 + a_{12}Y_2 &= -y_1(0) - G_1(s) \\ a_{21}Y_1 + (a_{22} - s)Y_2 &= -y_2(0) - G_2(s) \end{aligned}\right\}$$

Solving the system algebraically for $Y_1(s)$ and $Y_2(s)$ and then taking the inverse Laplace Transform we obtain the solution as: $y_1 = \mathcal{L}^{-1}(Y_1)$ and $y_2 = \mathcal{L}^{-1}(Y_2)$

Note: Setting $y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$, $g = \begin{pmatrix} g_1 \\ g_2 \end{pmatrix}$, $Y = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}$, $G = \begin{pmatrix} G_1 \\ G_2 \end{pmatrix}$ and $A = [a_{ij}]_{2\times 2}$, we can write the system of ODEs as: $y' = Ay + g$ and we obtain $(A - sI)Y = -y(0) - G$

Ex:



Tank T_1 initially contains 100 gallons of pure water. Tank T_2 initially contains 100 gallons of water in which 150 pounds of salt are dissolved. The inflow into T_1 is 2 gallons/min from T_2 and 6 gallons/min containing 6 pounds of salt from the outside. The inflow into T_2 is 8 gallons/min from T_1. The outflow from T_2 is $2+6=8$ gallons/min. The mixtures are kept uniform by stirring. Find and plot the salt contents $y_1(t)$ and $y_2(t)$ in T_1 and T_2 respectively.

- Time rate of change = Inflow/min − Outflow/min for the two tanks.

$$\left.\begin{aligned} y_1' &= -\frac{8}{100}y_1 + \frac{2}{100}y_2 + 6 \\ y_2' &= \frac{8}{100}y_1 - \frac{8}{100}y_2 \end{aligned}\right\} \quad \begin{aligned} y_1(0)&=0 \\ y_2(0)&=150 \end{aligned}$$

Taking Laplace Transform:

$$\left.\begin{aligned} (-.08-s)Y_1 + .02Y_1 &= -\frac{6}{s} \\ .08\,Y_1 + (-.08-s)Y_2 &= -150 \end{aligned}\right\} \quad \left(\begin{aligned} &\text{using } y_1(0)=0 \\ &\text{and } y_2(0)=150 \end{aligned}\right)$$

$$\Rightarrow \begin{pmatrix} -.08-s & .02 \\ .08 & -.08-s \end{pmatrix}\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \begin{pmatrix} -\frac{6}{s} \\ -150 \end{pmatrix}$$

$$\Rightarrow \quad Y_1 = \frac{\begin{vmatrix} -\frac{6}{s} & .02 \\ -150 & -.08-s \end{vmatrix}}{\begin{vmatrix} -.08-s & .02 \\ .08 & -.08-s \end{vmatrix}}$$

$$= \frac{-\frac{6}{s}(-.08-s) + 150(.02)}{(-.08-s)^2 - (.08)(.02)} = \frac{9 + \frac{.48}{s}}{s^2 + .16s + .0064 - .0016}$$

$$= \frac{9s + .48}{s(s^2 + .16s + .0048)} = \frac{9s + .48}{s(s+.12)(s+.04)}$$

$$\text{and} \quad Y_2 = \frac{\begin{vmatrix} -.08-s & -\frac{6}{s} \\ .08 & -150 \end{vmatrix}}{\begin{vmatrix} -.08-s & .02 \\ .08 & -.08-s \end{vmatrix}} = \frac{12 + 150s + \frac{.48}{s}}{(s+.12)(s+.04)}$$

$$= \frac{150s^2 + 12s + .48}{s\,(s+.12)(s+.04)}$$

Using Partial fraction decomposition:

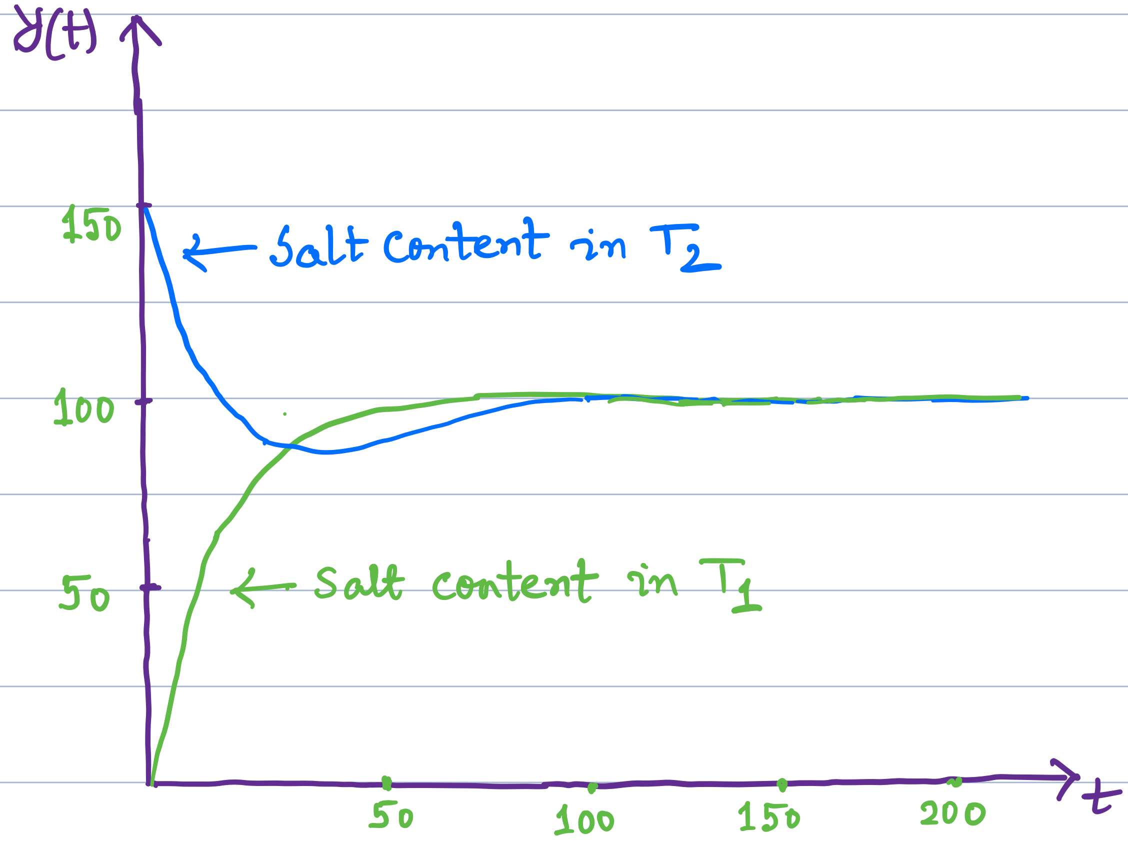
$$Y_1 = \frac{100}{s} - \frac{62.5}{s+.12} - \frac{37.5}{s+.04}$$

and $$Y_2 = \frac{100}{s} + \frac{125}{s+.12} - \frac{75}{s+.04}$$

By taking inverse Laplace Transform:

$$y_1 = 100 - 62.5\,e^{-.12t} - 37.5\,e^{-.04t}$$

and $$y_2 = 100 + 125\,e^{-.12t} - 75\,e^{-.04t}$$

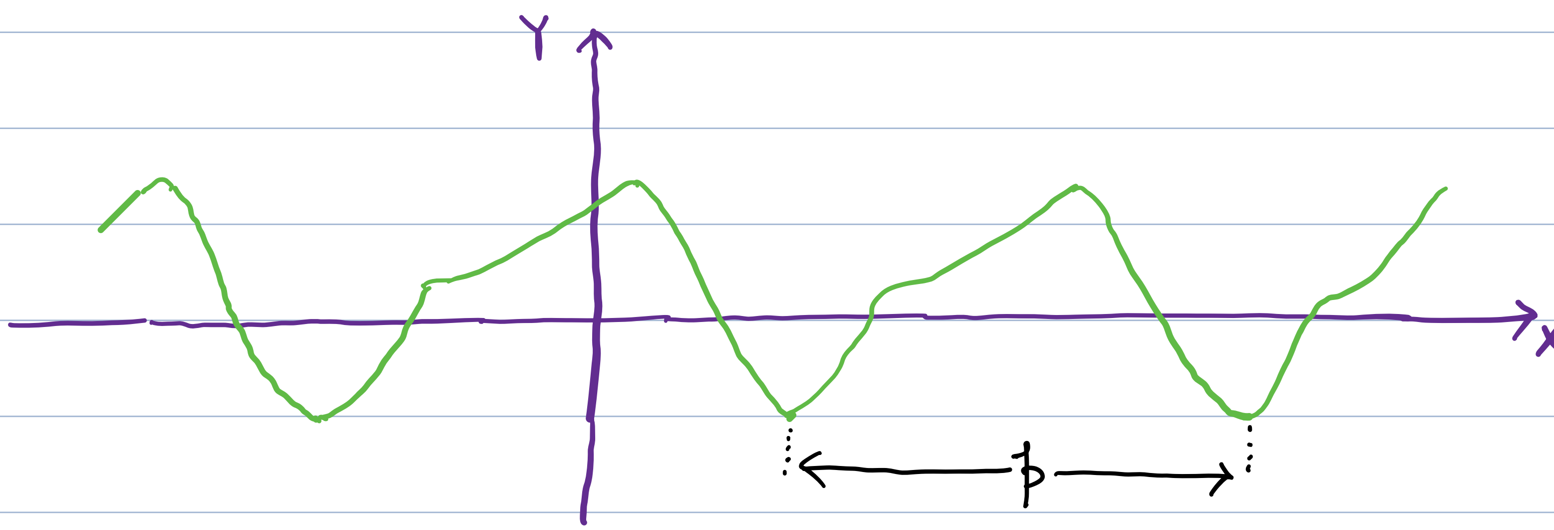


Fourier Series:

Trigonometric Series: It is a linear combination of $\{1, \cos x, \sin x, \cos(2x), \sin(2x), \ldots\}$ with coefficients $a_0, a_1, b_1, a_2, b_2, \ldots$ respectively.

Periodic Function: A function $f(x)$ is called a periodic function if $f(x)$ is defined for all real x, except possibly at some points, and if there is some positive number p, called a period of $f(x)$ such that

$$f(x+p) = f(x) \quad \text{for all } x \text{ in the domain of } f.$$



- The graph of a periodic function can be obtained by periodic repetition of its graph of any interval of length p.

- The smallest positive period is called the Fundamental period.

  If $f(x)$ has a period p, it also has a period $2p, 3p, \ldots, np$ for any integer $n = 1, 2, \ldots$

  $$f(x+2p) = f([x+p]+p) = f(x+p) = f(x)$$

  and $f(x+np) = f(x)$ for all x in the domain of f

- If $f(x)$ and $g(x)$ have period p, then $af(x) + bg(x)$ with any constants a and b also has the period p.

- $\sin x$, $\cos x$, $\tan x$, $\cot x$ are some familiar periodic function.

  Examples of functions that are <u>not</u> <u>periodic</u> : $\begin{cases} x, x^2, x^3, e^x, \cosh x, \sinh x, \\ \ln x \quad \text{etc.} \end{cases}$

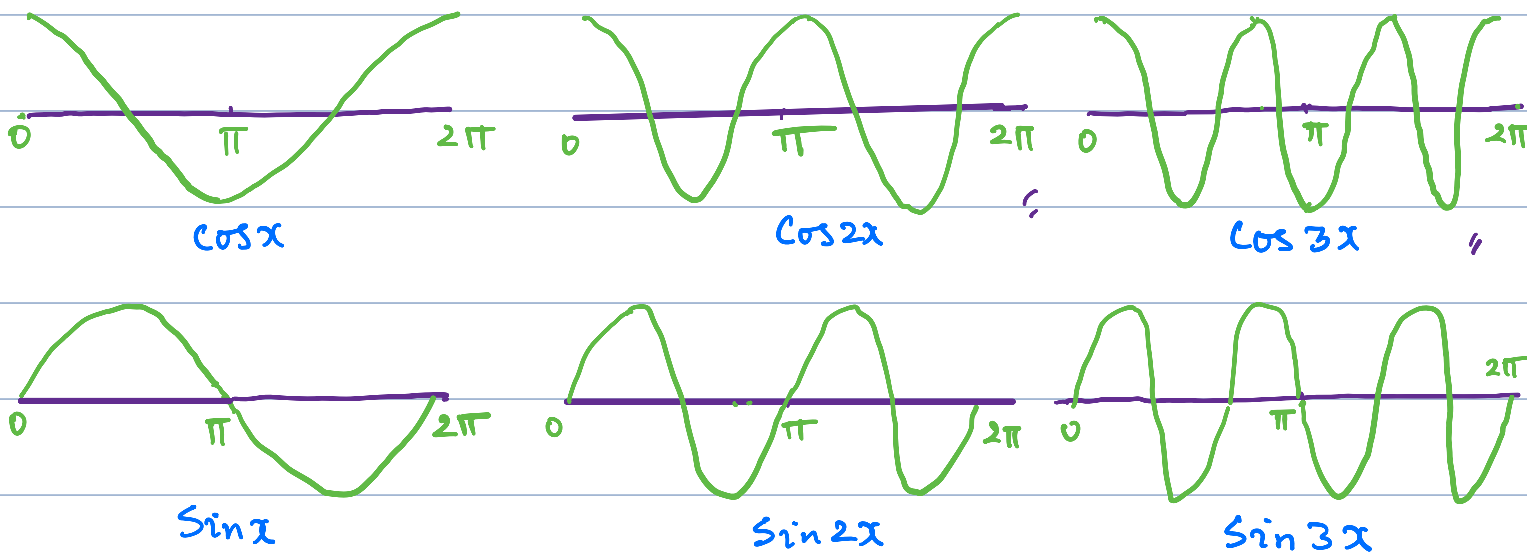
- $f(x) = \tan x$ is a periodic function that is not defined for all real x but undefined at countably many points

  $$x = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots\ldots$$

- In Fourier Series we consider representation of various functions of period 2π in terms of simple functions

$$1, \cos x, \sin x, \cos 2x, \sin 2x, \dots, \cos nx, \sin nx,$$

All these function have period 2π.
(1 is of course periodic with any period)
These functions form "Trigonometric System".



$\sin 2x, \cos 2x$ have fundamental period π. $\sin nx, \cos nx$ have fundamental period $\frac{2\pi}{n}$.

- A series of the form

$$a_0 + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \dots$$
$$= a_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$$

where $a_0, a_1, b_1, a_2, b_2, \dots$ are constants is called a Trigonometric Series.

Each term has period 2π. Hence if the coefficients $a_0, a_1, b_1, a_2, b_2, \ldots$ are such that the series converges, its sum will be a function of period 2π.

- Next let $f(x)$ be a function of period 2π and is such that it can be represented by a series $a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$

(i.e. the series converges and has sum $f(x)$)

then we can write

$$f(x) = a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$$

and the series $f(x)$ is called the Fourier series representation of $f(x)$.

The coefficients of the series are called Fourier Coefficients of $f(x)$

and they are given by the Euler's Formulas:

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx \text{ for } n=1,2,\ldots$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx \text{ for } n=1,2,\ldots$$

(Proof will be given later)

Note: There are functions $f(x)$ such that the series

$$a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$$

with $a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$, $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$

and $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$

doesnot converge or doesnot converge to $f(x)$ (i.e. the series has a sum different from $f(x)$)

In that case we will call the series $a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$

Fourier series corresponding to $f(x)$ (not the representation of $f(x)$)

and write

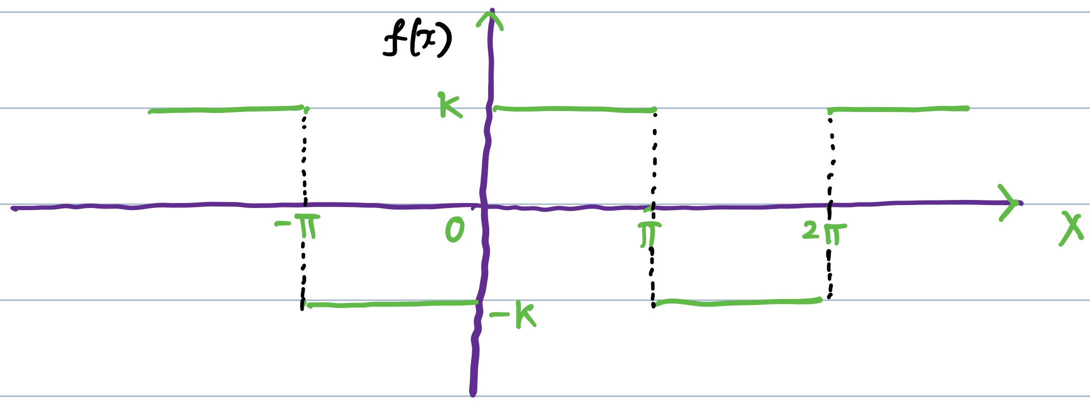
$$f \sim a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$$

Ex:

$$f(x) = \left.\begin{array}{lll} -K & \text{for} & -\pi < x < 0 \\ K & \text{for} & 0 < x < \pi \end{array}\right\}$$

and $f(x + 2\pi) = f(x)$

(Periodic rectangular wave)



$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{2\pi}\left[\int_{-\pi}^{0}(-K)\,dx + \int_{0}^{\pi}(K)\,dx\right]$$

$$= \frac{1}{2\pi}\left[-K\pi + K\pi\right] = 0$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0}(-K)\cos(nx)\,dx + \int_{0}^{\pi} K\cos(nx)\,dx\right]$$

$$= \frac{1}{\pi}\left[-K\,\frac{\sin(nx)}{n}\Bigg|_{-\pi}^{0} + K\,\frac{\sin nx}{n}\Bigg|_{0}^{\pi}\right]$$

$$= \frac{1}{\pi}(0) = 0$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,\sin(nx)\,dx$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0}(-k)\,\sin(nx)\,dx + \int_{0}^{\pi}(k)\,\sin(nx)\,dx\right]$$

$$= \frac{1}{\pi}\left[k\,\frac{\cos nx}{n}\Big|_{-\pi}^{0} - k\,\frac{\cos nx}{n}\Big|_{0}^{\pi}\right]$$

$$= \frac{k}{n\pi}\left[\cos 0 - \cos(-n\pi) - \cos(n\pi) + \cos 0\right]$$

$$= \frac{k}{n\pi}\left[2\cos 0 - 2\cos n\pi\right]$$

$$= \frac{2k}{n\pi}\left[1-(-1)^n\right]$$

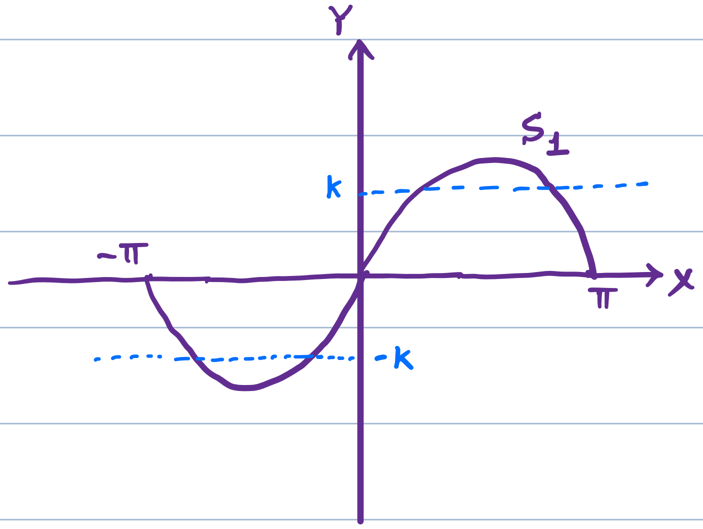
Hence $b_1 = \frac{4k}{\pi}$, $b_2 = 0$, $b_3 = \frac{4k}{3\pi}$, $b_4 = 0$,

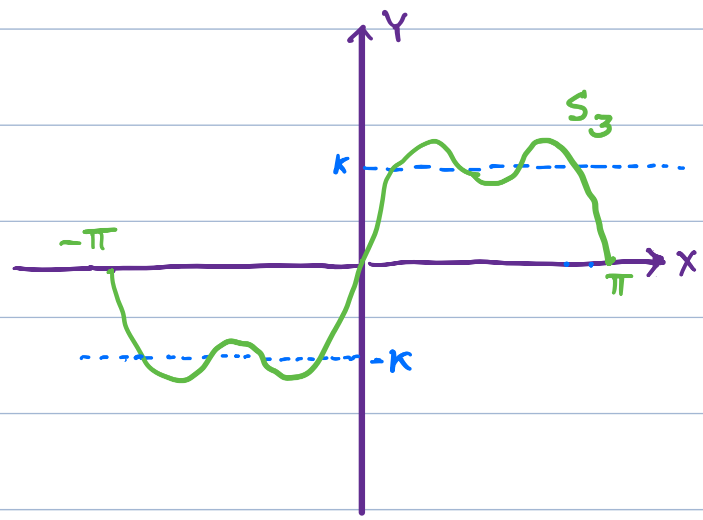
$b_5 = \frac{4k}{5\pi}$ . . .

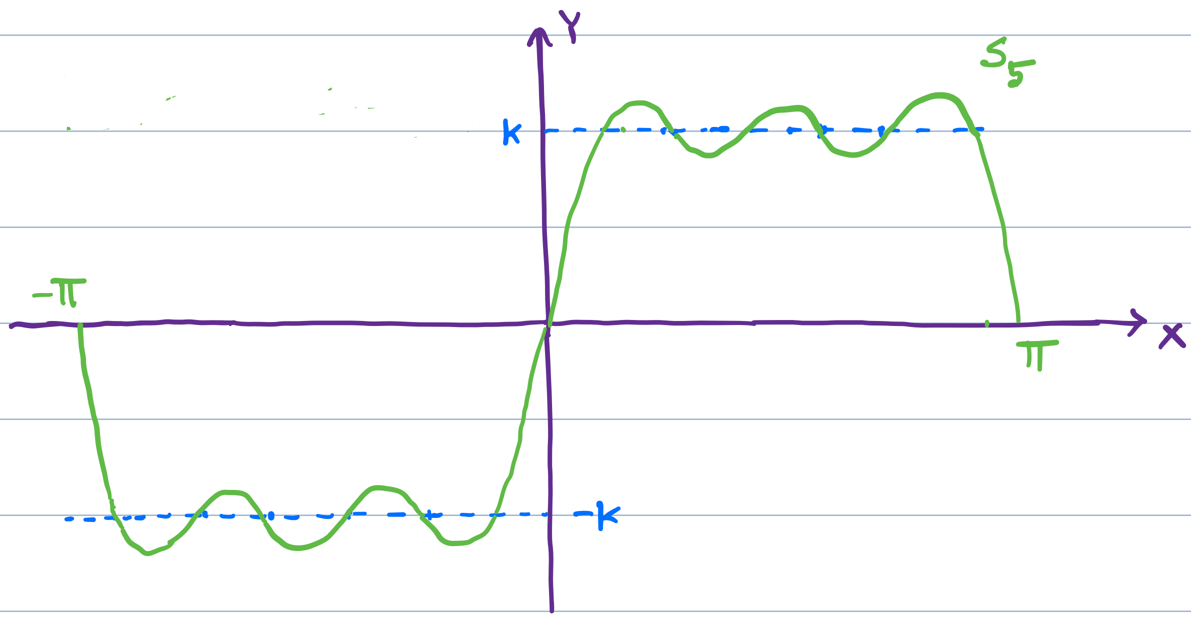
So, $$\boxed{f(x) = \sum_{n=1}^{\infty}\frac{2k}{n\pi}\left[1-(-1)^n\right]\sin nx}$$

Now $S_1 = \frac{4k}{\pi}\sin x$, $S_3 = \frac{4k}{\pi}\sin x + \frac{4k}{3\pi}\sin(3x)$

$$S_5 = \frac{4k}{\pi}\sin x + \frac{4k}{3\pi}\sin(3x) + \frac{4k}{5\pi}\sin(5x)$$







<u>Note:</u> • In the above example $f(x)$ is discontinuous at $x=0$. However the Fourier series is 0 at $x=0$

Note that $f(0+) = k$, $f(0-) = -k$

and $\dfrac{f(0+) + f(0-)}{2} = 0 =$ Value of the Fourier series at $x=0$

• Now $f(x) = \dfrac{4k}{\pi}\left(\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \cdots\right)$

Let us substitute $x = \dfrac{\pi}{2}$

Then $$k = \frac{4k}{\pi}\left(\sin\frac{\pi}{2} + \frac{1}{3}\sin\frac{3\pi}{2} + \frac{1}{5}\sin\frac{5\pi}{2} + \cdots\right)$$

$$\Rightarrow \quad k = \frac{4k}{\pi}\left(1 + \frac{1}{3}(-1) + \frac{1}{5}(1) - \cdots\cdots\right)$$

$$\Rightarrow \quad 1 = \frac{4}{\pi}\left(1 - \frac{1}{3} + \frac{1}{5} - \cdots\cdots\right)$$

$$\Rightarrow \quad \boxed{1 - \frac{1}{3} + \frac{1}{5} - \cdots\cdots = \frac{\pi}{4}}$$

Orthogonality :

Let V be the vector space of 2π-periodic functions. Then V is an inner product space with respect to the inner product :

$$\langle f, g\rangle = \int_{-\pi}^{\pi} f(x)\,g(x)\,dx$$

(It is the same inner product as in $L^2[-\pi, \pi]$)

In this space $\{1, \cos x, \sin x, \cos(2x), \sin(2x) \,\ldots\ldots\}$ is an "orthogonal basis".

$$\langle 1, \cos(nx)\rangle = \int_{-\pi}^{\pi} \cos(nx)\,dx = \left.\frac{\sin(nx)}{n}\right|_{-\pi}^{\pi} = 0$$

$$\langle 1, \sin(nx)\rangle = \int_{-\pi}^{\pi} \sin(nx) = -\left.\frac{\cos(nx)}{n}\right|_{-\pi}^{\pi} = 0$$

$$\langle \sin(nx), \cos(mx)\rangle = \int_{-\pi}^{\pi} \sin(nx)\cos(nx)\,dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi}\left[\sin((n+m)x) + \sin((n-m)x)\right]dx = 0$$

$$\langle \sin(nx), \sin(mx)\rangle = \int_{-\pi}^{\pi} \sin(nx)\sin(mx)\,dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi}\left[\cos((n-m)x) - \cos((n+m)x)\right]dx$$

$$\left.\begin{aligned} &= 0 \quad \text{if } n \neq m \\ \text{and } &= \frac{1}{2}\int_{-\pi}^{\pi}\left[1-\cos((2m)x)\right]dx = \pi \quad \text{if } n = m \end{aligned}\right\}$$

$$\langle \cos(nx), \cos(mx)\rangle = \int_{-\pi}^{\pi} \cos(nx)\cos(mx)\,dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi}\left[\cos((n+m)x) + \cos((n-m)x)\right]dx$$

$$\left.\begin{aligned} &= 0 \quad \text{if } n \neq m \\ &= \frac{1}{2}\int_{-\pi}^{\pi}\left[\cos((2m)x) + 1\right]dx = \pi \quad \text{if } n = m \end{aligned}\right\}$$

Note: The above set is not orthonormal.

$$\langle 1, 1\rangle = \|1\|^2 = 2\pi$$

$$\langle \cos(nx), \cos(nx)\rangle = \|\cos(nx)\|^2 = \pi$$

$$\langle \sin(nx), \sin(mx)\rangle = \|\sin(nx)\|^2 = \pi$$